

Roman Huts

647-XXX-XXXX | ___@gmail.com | [linkedin.com/in/romanh/](https://www.linkedin.com/in/romanh/) | github.com/rhuts | github.com/rhutsAMD

SUMMARY

Senior Software Engineer specializing in C++, GPU acceleration, video processing, and cross-platform SDK development. Experienced in shipping developer-facing technologies across Windows, Linux, and Android, including AMD AMF, AV1 integration into OBS, and AI-powered video processing.

EXPERIENCE

Senior Software Engineer, Developer Solutions

2022 – Present

AMD

Markham, ON / Remote

- Delivered **14** public [AMF SDK](#) releases, resolved **80+** **external developer issues**, and hundreds of internal bugs.
- Improved encoding efficiency by 30% for AMD GPU users by integrating AMD's **AV1 encoder** into [OBS](#) with around **20 million users** (C/C++).
- Developed **Flatpak** runtime packaging to bring AMD AMF support to the [freedesktop.org runtime](#) ecosystem, enabling AMD GPU-accelerated video encoding for Linux applications and reaching **679K+ monthly downloads** on Flathub.
- Designed and implemented **AI model compression and DLL embedding** to enable efficient distribution of an AMD AI video upscaling solution, and refactored the component architecture to support future extensibility.
- Designed and implemented **video rotation** support in the AMF VideoConverter, extending the SDK's video transformation capabilities.
- Implemented keyboard, touch, and game controller input support for AMD's open-source **Android streaming app**, demonstrating the capabilities of AMD's [Advanced Interactive Streaming SDK](#).

Software Development Engineer 2, Developer Solutions

2020 – 2022

AMD

Markham, ON / Remote

- Improved Pre-Analysis and Pre-Processing component performance by up to **15×** by implementing **hardware-accelerated** versions of CPU-based algorithms (**OpenCL / DX11 / DX12 / Vulkan / C++**) in the AMF SDK, improving the user experience for up to **285 million users** of AMD GPU drivers worldwide.
- Developed ONNX model-based audio processing demos for external customers, added Linux support, and integrated reusable components and plugins into the AMF SDK and internal applications.

Software Developer Intern, Game Engine Integration

2018 – 2019

AMD

Markham, ON

- Improved SDK robustness by fixing over 20 bugs for AMF and LiquidVR SDKs such as eGPU support (**C++**).
- Expanded support for AMD products by integrating **FreeSync 2 HDR** into **Unreal Engine 4** (**C++**).
- Owned the team's Jenkins CI/CD farm, maintaining **98%** uptime and improving automation reliability (**Python / Bash / Groovy**).

BUSINESS PROJECTS

NDIO Customs | *Node, Javascript, electronics, lithium battery systems*

- Two-time **1st place winner** for Special Construction class custom build at the [North American International Motorcycle Supershow](#)(2023 & 2025).

Music Licensing Lead Generation (DCSIL Program) | *Android Studio, Java, Flask, Python, Heroku, Redis, MongoDB*

- Lead team development of cross platform app for generating non-confrontational leads.
- Pitched to **multiple leaders** in the **music licensing** industry space such as BMAT USA and Dataclef (subsidiary of SOCAN). ([MVP Tech Demo](#))

AWARDS

- CG Recognition award for shipping **AV1** with **OBS** (AMD 2022)
- Spotlight award for role in **perforce to git migration** (AMD 2021)
- Recognition award for **outstanding** co-op **mentorship** (AMD 2021)

TECHNICAL SKILLS

Languages: C++, C, Python, Bash, JavaScript, SQL, HTML, CSS

Computer Graphics: OpenCL, DirectX11, Vulkan, DirectX12

Developer Tools: CMake, Make, Visual Studio, Git, GitHub, Perforce, JIRA, Confluence, GDB, GPUView, Valgrind, RadeonGPU Analyzer, Elecard StreamEye Stream Analyzer, ONNX Runtime, Ultralytics, OpenCV, CodeQL,

Operating Systems: Linux, Windows

EDUCATION

University of Toronto

Honors Bachelor of Science with Distinction, Computer Science Specialist

Toronto, ON

2015 – 2020